MILITARY SPECIFICATION

TEST SET, CONTROL PANEL, TS-2657/TSM-100A

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of Control Panel Test Set, hereinafter referred to as the "test set", which forms a part of the Electronics System Test Set type AN/TSM-100A.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitations.

SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	-	Parts, Equipment and Tools for Army Materiel; Packaging and Packing of
MIL-T-43435	-	Tape: Impregnated, Lacing and Tying
MIL-C-46357	-	Computer, Gun Direction; M18 Manufacturing Processes and Inspection of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-454	-	Standard General Requirements for Electronic Equipment
MIL-STD-810	-	Environmental Test Methods

2.1.2 Other Government documents, drawings, and publications. The following other Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.4)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

11748027	-	Panel Assembly, Control Test Unit
12000197	-	Schematic Diagram, Electrical, Control Panel, Test Unit

PACKAGING DATA SHEET

11748027	-	Packaging of Test Set, Control Panel

(Copies of specification, standards drawings and packaging data sheets required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Fabrication. The test set shall be manufactured in accordance with Drawing 11748027 and drawings pertaining thereto. The subassemblies of the test set shall have been inspected and accepted in accordance with their respective drawings or specifications prior to their installation into the test set.

3.2 General specification. Where applicable, the test set shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

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3.2.1 MIL-F-13926.

a. Dimensions and Tolerances

b. Inorganic protective surface finishes

c. Workmanship

3.2.2 MIL-C-46357.

a. Wiring. The requirements for wiring of MIL-C-46357 shall apply with modification as follows:

(1) Spot ties. Tape; Type 1, Finish B per MIL-T-43435 may be utilized without the need for varnish on the knots.

(2) Wire termination. Removal of insulation by controlled heat methods shall be acceptable.

(3) Chassis cleaning. Compressed air or vacuum may be utilized for-cleaning. Vibration shall not be used to clean the chassis.

b. Threaded fasteners. The requirements for threaded fasteners of MIL-C-46357 shall apply with modification as follows:

(1) Thread projection. Maximum projection of screws shall be that of the next standard length screw beyond the minimum projection, unless electrical or mechanical interference exists.

c. Marking. The requirements for marking of MIL-C-46357 shall apply with modification as follows:

(1) Etching and engraving. Solvo photo etch process is acceptable.

3.3 Environmental service conditions.

3.3.1 Vibration (non-operating). The test set shall meet the requirement of 3.4 after being subjected to continuous vibration for two hours along each of three mutually perpendicular axes with in the frequency range and amplitude as follows:

Frequency (Hz ± 2 percent)	Excursion (double amplitude) (inches ±10 percent)
5 to 15	0.06
15 to 25	0.04
25 to 55	0.02

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The rate of change of frequency at each frequency range shall not exceed one cycle per second per second (1 cps/sec). Vibration tests shall be conducted under ambient room conditions.

3.3.2 Temperatures.

3.3.2.1 Operating. The test set shall meet the requirements of 3.4 during exposure at any temperature between +32°F and +131°F and after it is thermally stabilized at both low and high ambient temperature extremes.

3.3.2.2 Non-operating (storage). The test set shall meet the requirements of 3.4 at a temperature of between +60°F and +90°F (room ambient) after having been exposed to constant temperature extremes of -60°F for a period of 24 hours and +160°F for a period of 48 hours (after temperature stabilization).

3.3.3 Humidity (non-operating). The test set shall meet the requirements of 3.4 and shall not suffer any damage or deterioration which shall impair satisfactory performance after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.3.4 Shock. The test set shall function as specified in 3.4 and shall show no evidence of damage or be adversely affected atter being subjected to a total of 18 impact shocks consisting of three shocks in opposing directions along each of the three mutually perpendicular axes. Each shock shall have an acceleration of 15 gravity units (G's) and a pulse duration of 11 + 1 millisecond. The G value of each inpulse shall be within ±10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5.5 milliseconds.

3.3.5 Altitude.

3.3.5.1 Operating. The test set shall not be damaged or adversely affected when exposed to barometric pressure encountered at an altitude of 10,000 feet above sea level (20.6 inches Hg). During and subsequent to such exposure the test set shall meet the requirement of 3.4.

3.3.5.2 Non-operating. The test set shall meet the requirements of 3.4 after exposure to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg) for a period of not less than 10 minutes. The test set shall not be damaged or be adversely affected due to such exposure.

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3.4 Electrical.

3.4.1 Electrical continuity. The circuitry shall be in accordance with electrical schematic Drawing 12000197. The symbols referenced in subsequent paragraphs pertain to this schematic and are used therein.

3.4.2 Functional.

3.4.2.1 Continuity Check and Switch Functions check. With the SWITCH FUNCTION CHECK switch and the CONTINUITY CHECK switch in the positions specified in Table I, the resistance between TP5 and the specified connector pins (Table I) shall be less than 0.5 ohms.

TABLE I. Switch test.

SWITCH FUNCTION CHECK Switch Position	CONTINUITY CHECK Switch Position	Connectors & Pin(s)
A	1	J23-B, J24-A
A	2	J24-G, K
A	3	J24-C, L
A	4	J22-BK
A	5	J22-BT, BM
A	6	J22-BU, BP
A	7	J22-BS, BV
A	8	J22-BA
A	9	J22-AZ, AP
A	OFF	J22-AR, BC
B	1	J22-AV
B	2	J23-X, W J22-AX, AL, AU AC, BJ, i, k
B	3	J22-AY
B	4	J22-AD
B	5	J22-AH

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TABLE I. Switch test (Cont'd).

SWITCH FUNCTION CHECK Switch Position	CONTINUITY CHECK Switch Position	Connectors & Pin(s)
B	6	J22-BL, BN, u, t, J23-V J114-A
B	7	J22-AW, BB
B	8	J22-j
B	9	J22-w, AE, y, z
B	OFF	J22-x, AJ
C	1	J22-m, q
C	2	J22-AF
C	3	J22-AT, BD
C	4	J22-v
C	5	J22-BR
C	6	J22-AA
C	7	J22-d
C	8	J22-h, c
C	9	J23-K, Z
C	OFF	J23-n, D
D	1	J23-R, P
D	2	J23-k, N
D	3	J23-c
D	4	J23-s, A
D	5	J23-a, b
D	6	J22-AK

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TABLE I. Switch test (Cont'd).

SWITCH FUNCTION CHECK Switch Position	CONTINUITY CHECK Switch Position	Connectors & Pin(s)
D	7	J22-AS
D	8	J22-p
D	9	J23-m, e
D	OFF	J23-h, j
E	1	J23-d, E
E	2	J23-r
E	3	J23-q, C
E	4	J31-A
E	5	J31-L
E	6	J31-N
E	7	J31-S
E	8	J31-E
E	9	J31-J
E	OFF	J31-R
F	1	J31-U
F	2	J31-C
F	3	J31-T
F	4	J31-B
F	5	J31-X
F	6	J31-Y
F	7	J31-Z
F	8	J31-W
F	9	J31-V

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TABLE I. Switch test (Cont'd).

SWITCH FUNCTION CHECK Switch Position	CONTINUITY CHECK Switch Position	Connectors & Pin(s)
F	OFF	J31-c
G	1	J31-G
G	2	J31-b
G	3	J31-M
G	4	J31-K
G	5	J31-P
G	6	J31-F
G	7	J31-H
G	8	J31-D
G	9	J31-a

3.4.2.2 SWITCH FUNCTION CHECK indicator. With +22.0 + 0.1 VDC applied between the specified pin in Table II and GROUND, the SWITCH-FUNCTION CHECK indicator shall light only when the SWITCH FUNCTION CHECK switch is in the position specified in Table II.

TABLE II. Switch test.

SWITCH FUNCTION CHECK switch position	+22.0 ± 0.1 VDC to connector and pins
A	J23-r
B	J23-p, J22-BE
C	J23-d
D	J23-m
E	J22-AM
F	J23-a
G	J22-J
H	J22-AF
I	J23-c

3.4.2.3 CONTINUITY CHECK indicator. With +22.0 ± 0.1 VDC applied between the specified pin in Table III and GROUND, the CONTINUITY CHECK indicator shall light only when the CONTINUITY CHECK switch is in the position specified in Table III

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TABLE III. Switch test.

CONTINUITY CHECK switch position	+22.0 ± 0.1 VDC to connector and pins
1	J22-BK
2	J22-BT
3	J22-BU
4	J22-BS
5	J22-AZ
6	J22-AR
7	J22-BD
8	J22-a, g
9	J22-h

3.4.2.4 POTENTIOMETER SELECT switch.

3.4.2.4.1 TEST POINT NO. 1 (TP1) continuity. The resistance between TP1 and the connector pins with the POTENTIOMETER SELECT switch in the corresponding positions shall be less than 0.5 ohms as specified in Table IV.

TABLE IV. Switch test.

POTENTIOMETER CHECK switch position	connector and pin
1	J23-f
2	J23-f
3	J24-F
4	J22-AN
5	J22-f
6	J22-n

3.4.2.4.2 TEST POINT NO. 2 (TP2) continuity. The resistance between TP2 and the connector pins with the POTENTIOMETER SELECT switch in the corresponding positions shall be less than 0.5 ohms as specified in Table V.

TABLE V. Switch test.

POTENTIOMETER CHECK switch position	connector and pin
1	J23-U
2	J23-U
3	J24-D
4	J22-AB
5	J22-s
6	J22-r

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3.4.2.4.3 TEST POINT WIPER (TP3). The resistance between WIPER (TP3) and the connector pins with the POTENTIOMETER SELECT switch in the corresponding position shall be less than 0.5 ohms as specified in Table VI.

TABLE VI. Switch test.

POTENTIOMETER CHECK switch position	connector and pins
1	J23-S
2	J23-T
3	J23-P
4	J22-AM
5	J22-g

3.4.2.5 RC network.

3.4.2.5.1 Conditions of operation. The requirements of 3.4.2.5 shall be met with the conditions listed below.

a. +24.0 + 0.5 VDC (600 milliamperes or greater) source applied between J114-C (+) and J114-A (-).

b. A 1K ohm ±1 percent, 1 watt, load applied between JZ2-AY and J22-BL.

c. POWER ON/OFF switch in the "ON" position. The power +24 VDC indicator lamp shall be LIT.

3.4.2.5.2 RC charging. The waveform shown in Figure 1 shall be obtained across terminals E3 (+) and E4 when the POTENTIOMETER SELECT switch is placed from position 6 to position 7.

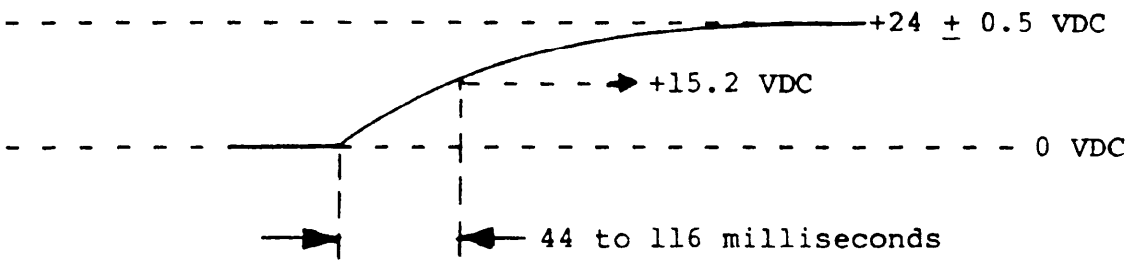


FIGURE 1. RC charging curve.

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3.4.2 .5.3 RC discharging. The wave form shown in Figure 2 shall be obtained across terminals E1 (+) and E2 when the POTENTIOMETER SELECT switch is switched to POSITION 6 from POSITION 7. The waveform shown in Figure 2 shall also be obtained across J22-AY (+) and J22-BL when the POTENTIOMETER SELECT switch is switched to POSITION 8 from POSITION 7.

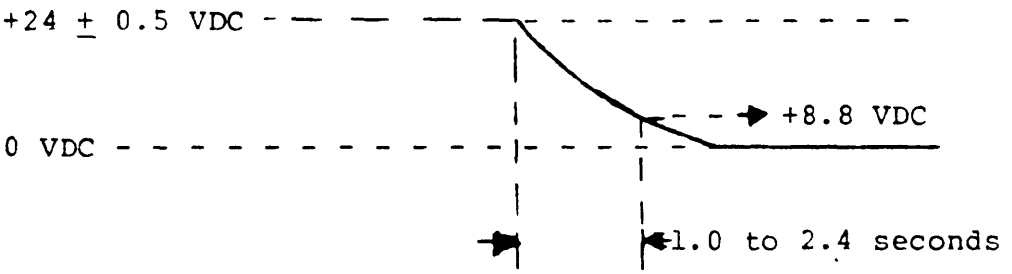


FIGURE 2. RC discharging curve.

6 +15V Adjust regulator. With +24 + 0.5 VDC applied between J114-C (+) and J114-A (-) and a resistor of 1K ± 1% between J23f and J23U, the voltage at J23-f shall be capable of being adjusted to 15 ± 0.1 VDC. After a minimum of 15 minutes the voltage output should be 15 ± 0.5 VDC.

3.5 First article. Unless otherwise specified in the contract or purchase order, a sample should be subjected to first article inspection (see 4.2 and 6.2).

3.6 Workmanship. The requirements of MIL-STD-454 and other applicable specifications shall apply.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall

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inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of asuring that all products or supplies submitted to the Government for acceptance comply with all requirements of trhe contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 First article (initial production) approval. The requirements for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall be 1 test set manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sampler including certificates of conformance for materials.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor ‘s inspection.

4.3 Inspection provisions.

4.3.1 Submission of product. Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.3.2 Examinations and tests. Except as otherwise specified herein, components and subassemblies shall be inspected in accordance with the “Inspection provisions for components and subassemblies” and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Table VII and VIII and methods of inspection in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

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4.3.2.1 Components. All components shall be inspected to the applicable requirements of MIL-F-13926 and MIL-C-46357.

4.3.2.2 Control tests. One test set shall be selected at random as a control sample from each 100 produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests of Table VII after they have met the requirements and tests of Table VIII.

TABLE VII. Control test.

Test	Requirement	Test Procedures
Fabrication	3.1	Applicable drawings - visual
General specification	3.2	MIL-F-13926/MIL-C-46357 - visual
Vibration	3.3.1	4.5.1.1, 4.5.1.2
Temperatures	3.3.2	4.5.1.1, 4.5.1.3
Humidity	3.3.3	4.5.1.1, 4.5.1.4
Shock	3.3.4	4.5.1.1, 4.5.1.5
Altitude	3.3.5	4.5.1.1, 4.5.1.6

4.3.3 Rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3.4 Functional tests. The requirements and tests in Table VIII shall be inspected on a 100% basis. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a. Temperature +50° to +90°F

b. Relative humidity 90% RH max

c. Pressure 28 to 32 inches of Hg

TABLE VIII. Functional test. (100 percent inspection)

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Characteristic		Requirement	Procedure
101	Continuity test	3.4.1	4.5.2.1
102	Continuity Check and Switch Functions check	3.4.2.1	4.5.2.2.1
103	SWITCH FUNCTION CHECK indicator	3.4.2.2	4.5.2.2.2
104	CONTINUITY CHECK indicator	3.4.2.3	4.5.2.2.3
105	TEST POINT NO. 1 (TP 1) Continuity	3.4.2.4.1	4.5.2.2.4
106	TEST POINT NO. 2 (TP 2) Continuity	3.4.2.4.2	4.5.2.2.5
107	TEST POINT WIPER (TP 3) Continuity	3.4.2.4.3	4.5.2.2.6
108	Conditions of operation	3.4.2.5.1	4.5.2.2.7
109	RC charging	3.4.2.5.2	4.5.2.2.7
110	RC discharging	3.4.2.5.3	4.5.2.2.8
111	+15V Adjust Regulator	3.4.2.6	4.5.2.2.9

4.4 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements herein. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design. The contractor shall provide inspection equipment compatible with the “Methods of inspection” specified in 4.5 and with the "Component Inspection” and "Test Facilities” requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.1.1 Inspection equipment. In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5:

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TABLE IX. Inspection equipment list.

Item	Description
1. OHMMETER	0 to 1000 ohms. Accuracy and resolution 1% or better.
2. DIGITAL VOLTMETER	A range of at least .0010 to 999.9 VDC with four significant figures. Accuracy ± 0.2% of reading. Input impedence 10 megohms or better.
3. POWER SUPPLY	0-30 VDC and 0-600 MA, current and voltage adjustable, regulated to ± 1% or better. Less than 0.1% ripple.
4. OSCILLOSCOPE	DC-15 MHZ; input impedence 10 megohms or greater, 5 volt/CM sensitivity; internal sweep; 20 millisecond/CM and 500 millisecond sweep speeds.
5. RESISTOR	1K ohm ± 1%, 1 watt wire wound.
6. RESISTOR	1K ohm ± 5%, 1/2 watt

4.5 Methods of inspection.

4.5.1 Environmental.

4.5.1.1 Test conditions. The test set shall be mounted in an upright position in a manner similar to that in which it will be in service and shall be subjected to the environmental tests of the following subparagraphs. Except for the high and low temperature tests or the humidity tests, as applicable, the temperature and humidity shall be room ambient.

4.5.1.2 Vibration. The test set shall be mounted on the vibration table in normal service position. The test shall be performed as specified in Method 514, Procedure I, Category 1, of MIL-STD-810 except as modified by paragraph 3.3.1 herein. After vibration the test set shall then be tested to determine compliance with requirements specified under 3.3.1.

4.5.1.3 Temperature. Test in accordance with Procedure I and II of Method 501 in MIL-STD-810 for high temperature operating and non-operating under conditions specified in 3.3.2. Test in accordance with procedure I and II of Method 502 in MIL-STD-810 for low temperature operating and non-operating under conditions specified in 3.3.2 and verify requirements of 3.3.2.1 and 3.3.2.2.

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4.5.1.4 Humidity. The humidity test shall be performed in accordance with Procedure III of Method 507 in MIL-STD-810, except that in Step 7 the operational checkout shall be performed at the end of the tenth cycle only and at standard ambient conditions as specified in 4.5.2 for compliance with 3.3.3.

4.5.1.5 Shock. Test in accordance with Procedure I of Method 516 of MIL-STD-810 except that shock pulses shall be specified in 3.3.4. After the shock impulses the test set shall be tested as specified in 4.5.2 for compliance with 3.3.4.

4.5.1.6 Altitude (operating and non-operating). The altitude test shall be performed in accordance with Procedures II and 111 of Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.5.2 for compliance with 3.3.5.

4.5.2 Electrical test.

4.5.2.1 Continuity test. A continuity test shall be performed by using an ohmmeter item 1 of 4.4.1.1 to verify conformance to 3.4.1.

4.5.2.2 Functional tests.

4.5.2.2.1 Continuity Check and Switch Function Check. Using the ohmmeter (item 1 of 4.4.1.1) check the resistance between TP5 and the various connector pins, Table I, to verify the requirements of 3.4.2.1.

4.5.2.2.2 Switch function check indicator lamp test. Using the DVM, item 2 of 4.4.1.1, set the POWER SUPPLY, item 3 of 4.4.1.1, to +22.0 + 0.1VDC. Connect the POWER SUPPLY between each point specified in TABLE II one at a time, and ground. Operate the SWITCH FUNCTION CHECK switch through all positions for each connection and verify that the SWITCH FUNCTION CHECK lamp is “LIT” for each specified position to verify the requirements of 3.4.2.2. After test remove the power supply.

4.5.2.2.3 Continuity check indicator test. Using the DVM, item 2 of 4.4.1.1, set the POWER SUPPLY, item 3 of 4.4.1.1, to 22.0 + 0.1 VDC. Connect the POWER SUPPLY between each point specified in Table III one at a time, and GROUND. Operate the CONTINUITY CHECK rotary switch through all positions for each connection and verify that the CONTINUITY CHECK lamp is “LIT” for each specified position to verify the requirements of 3.4.2.3. After test remove the power supply.

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4.5.2.2.4 TP1 continuity. Using an OHMMETER (item 1 of 4.4.1.1) check the resistance between TP1 and the various connector pins as listed in Table IV to verify the requirements of 3.4.2.4.1.

4.5.2.2.5 TP2 continuity. Using an OHMMETER (item 1 of 4.4.1.1) check the resistnce between TP2 and the various connector pins as listed in Table V, to verify the requirements of 3.4.2.4.2.

4.5.2.2.6 Wiper (TP3) continuity. Using an OHMMETER (item 1 of 4.4.1.1) check the resistance between WIPER test point (TP3) and the various connector pins as listed in Table VI, to verify the requirements of 3.4.2.4.3.

4.5.2.2.7 RC charging. Place POTENTIOMETER SELECT switch to position 6 and the POWER switch to the OFF position. Apply +24.0 + 0.5 VDC between J114-C (+) and J114-A (-) and place the POWER switch to the ON position, the +24 VDC POWER lamp shall light. With the vertical input of the oscilloscope (item 4 of 4.4.1.1) connected to E3 and ground to E4, place the POTENTIOMETER SELECT switch to position 7 and verify the waveform of Figure I of 3.4.2.5.2. To discharge the RC circuit place the POTENTIOMETER select switch to position 6 for a minimum of 10 seconds and then back to position 7 to observe waveform of Figure I.

4.5.2.2.8 RC discharging.

4.5.2.2.8.1 Potentiometer Select switch position 6 discharging. Place the POTENTIOMETER SELECT switch to position 7 and the POWER switch to the ON position, the +24VDC power lamp shall light. With the vertical input of the oscilloscope (Item 4 of 4.4.1.1) to El and ground to E2, place the POTENTIOMETER SELECT switch to position 6 and verify the waveform of Figure 2 of 3.4.2.5.3. To charge the RC circuit place the POTENTIOMETER SELECT switch to position 7 (5 seconds minimum) and back to position 6 to observe the waveform of Figure 2.

4.5.2.2.8.2 Potentiometer Select switch position 8 discharging. Connect the 1K ohm resistor (item 5 of 4.4.1.1) between J22-AY and J22-BL. Place the POTENTIOMETER SELECT switch to position 7 and the POWER switch to the ON position, the +24 VDC power lamp shall light. With the vertical input of the oscilloscope (item 4 of 4.4.1.1) to J22-AY and ground to J22-BL, place the POTENTIOMETER SELECT switch to position 8 and verify the waveform of Figure 2 of 3.4.2.5.3. To charge the RC circuit place the POTENTIOMETER SELECT to position 7 (minimum of 5 seconds) and back to position 8 to observe the waveform of Figure 2. After test remove the 1K resistor.

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4.5.2.2.9 +15V Adjust regulator. Connect a 1K ohm 5% resister (item 6 of 4.4.1.1) between J23-f and J23-U. Apply +24 ± 0.5 VDC between J114-C (+) and J114-A (-) and place POWER switch to ON position, the +24 VDC power lamp shall lit. With voltmeter (item 2 of 4.4.1.1) measure the output across 1K ohm LOAD (J23-f (+) and J23-U) and turn +15 VDC ADJUST knob to obtain a reading of +15 + 0.1 VDC. After 15 minutes (minimum) the voltage at J23-F shall be +15 ± 0.5 VDC. After test remove all test components.

5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Packaging Data Sheet 11748027.

6. NOTES

6.1 Intended use. This test set is designed for rack-mounting into the AN/TSM-100A Electronics System Test Set, which is used for troubleshooting and performance checking of specific weapon system components.

6.2 Ordering data. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this and the item detail specifications.

b. Applicable stock number.

c. A block of serial numbers to identify each test set.

d. Quality Conformance Inspection, if other than specified in Section 4 of item detail specification.

e. First Article Sample requirements, if other than specified in Section 4 of item detail specification.

f. Packaging requirements, if other than specified in Section 5 of item detail specification.

g. Certificate of conformance for each lot or shipment of product.

6.2.2 Contract data requirements.

a. Contract data requirements for inspection equipment designs conforming to Data Item Description DI-R-1714, tailored, with appropriate addendum to each (see 6.3).

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b. Data Cards - When required by incorporation of DD Form 1423 into the contract, data cards shall be prepared for each lot in accordance with Data Item Description DI-L-1410.

6.3 Inspection equipment designs. Inspection equipment designs are of two types - Government designs and contractor designs. Government designs are designated by drawing numbers in the appropriate paragraph in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government design is provided; however, unless otherwise specified, the contractor may submit alternate or modified contractor designs in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 Government designs. Government designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in DOD-STD-100.

c. An envelope drawing as defined in DOD-DTS-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs. Contractor designs are required for all inspection equipment for which Government designs are not specified and include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with Government designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per DOD-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The item detail specification number, paragraph number and defect number from Section 4 shall be referenced on each contractor design together with the component or assembly drawing number, revision letter and date to which the specific design applies.

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6. 3.3 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements Lists (DD Form 1423) . Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAF-I(D), Dover, NJ 07801-5001.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by and identified as US Army Armament Research and Development Command, (ARRADCOM), Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.5 Changes to previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-60880A	TEST SET, CONTROL PANEL, TS-2657/TSM-100A

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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